Induction Heads Modulate Repeated-Token Bias but Are Not a Singular Source of Leakage in GPT-2 SMALL

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Abstract

Induction heads are one of the clearest mechanistic candidates for in-context copying, so they are a natural place to look for leakage-like behavior in language models. Our main question is whether empirically identified induction-like heads in GPT-2 SMALL push the model toward copying repeated prompt content even when the prompt instead asks for a diffuse or random answer. We score attention heads with an induction-head detector in TransformerLens, ablate the top-scoring heads by zeroing their `hook_z` activations, and compare this intervention against matched random-head ablations on three tasks: a controlled copy task, a random-choice leakage task, and a small WikiText utility check. The strongest final run identifies `L5H5` and `L5H1` as the top two induction-like heads. Ablating them reduces repeated-token bias on random-choice prompts from 0.0496 to 0.0211, but it also reduces normalized entropy from 0.860 to 0.819 and increases WikiText perplexity from 89.9 to 169.7. Sensitivity runs show that the conclusion is not stable across high-scoring head subsets: adding `L6H9` raises repeated-token bias instead of lowering it. These results show that induction-like heads matter for both intended and unintended copying, but they do not behave like a single large causal source of leakage in GPT-2 SMALL. Any mitigation built around these circuits must therefore be evaluated jointly for behavior, entropy, and utility.

1 Introduction

Large language models often repeat salient prompt content even when the task does not call for copying. That behavior matters in privacy settings, in robustness work on prompt contamination, and in applications that need controlled randomness rather than deterministic reuse of context. A mechanistic explanation would be valuable because it would turn an undesirable behavior into a concrete intervention target.

Induction heads are a strong candidate for that role. Prior work argues that induction heads implement a copy-and-complete algorithm that supports in-context learning Olsson et al. [2022], Crosbie and Shutova [2024], Ren et al. [2024]. More recent work links the same head family to repetitive generation failures Wang et al. [2025]. At the same time, work on memorization and extraction warns that leakage is often distributed and entangled with general language modeling ability rather than isolated in one small circuit Lee et al. [2022], Ippolito et al. [2022], Huang et al. [2024]. These two views create a clear empirical question: do induction heads materially drive unwanted copying pressure on prompts that should instead yield diffuse or random outputs?

This paper studies that question in GPT-2 SMALL. We first identify induction-like heads empirically using repeated-sequence prompts and a standard head detector. We then apply the same head-level intervention to two behaviors in the same model: intended copying on a controlled copy task and unintended copying on random-choice prompts with repeated distractor context. Figure 1 shows that the detector recovers a small set of mid-layer heads that align with the canonical induction-head pattern. The resulting ablations let us test whether removing those heads cleanly lowers repeated-token

bias, whether any change is specific relative to random-head controls, and whether the intervention preserves general utility.

The answer is mixed. In the strongest final run, ablating L5H5 and L5H1 lowers repeated-token bias by 0.0285 relative to the full model, with a 95% bootstrap confidence interval of $[-0.0400, -0.0172]$. However, the same intervention lowers normalized entropy by 0.0417 and raises WikiText perplexity from 89.9 to 169.7. More importantly, the causal story changes sign in sensitivity runs: when L6H9 is included among the ablated induction-like heads, repeated-token bias increases instead of decreasing. That sign flip is the central empirical result of the paper. It argues against the simple claim that induction heads are a single large source of leakage.

We make four contributions:

- We identify a compact set of induction-like heads in GPT-2 SMALL and verify that they occupy the expected mid-layer region.
- We conduct a head-level causal study that tests the same intervention on a controlled copy task, a random-choice leakage task, and a natural-text utility check.
- We show that the strongest ablation lowers repeated-token bias but also lowers entropy and substantially harms language modeling utility.
- We show that the bias effect reverses across high-scoring head subsets, which rules out a monolithic "ablate induction heads, remove leakage" explanation in this model.

The rest of the paper is organized as follows. Section 2 positions the study relative to induction-head, memorization, and randomness-control work. Section 3 describes the tasks, interventions, and evaluation metrics. Section 4 presents the main ablation results and sensitivity analyses. Section 5 discusses what these results imply for mechanistic mitigation and for future leakage studies.

2 Related Work

Induction heads and in-context learning. The modern induction-head literature treats these heads as a concrete circuit for copy-based pattern completion. Olsson et al. identify induction heads as a likely mechanistic driver of in-context learning and show strong causal evidence in small attention-only models Olsson et al. [2022]. Crosbie and Shutova extend that claim to larger modern LLMs, where ablating a small fraction of induction heads sharply reduces few-shot gains Crosbie and Shutova [2024]. Ren et al. broaden the picture further by identifying semantic induction heads that recover structured relations rather than only literal token matches Ren et al. [2024]. We build on this line of work by using empirical induction-head scoring to choose ablation targets in GPT-2 SMALL, but our target behavior is different: we focus on unwanted copying pressure in prompts that explicitly ask for a diffuse answer.

Induction heads and repetition pathologies. Wang et al. provide the closest prior hypothesis to ours. They argue that induction-head dominance mechanistically explains repetition curse in instruction-tuned LLMs and propose a softer descaling intervention rather than hard ablation Wang et al. [2025]. Their results suggest that induction heads can hurt output diversity in repetitive settings. Our work asks a narrower but complementary question: whether induction-like heads also push probability mass toward a repeated distractor when the task is to choose at random from a constrained set. Unlike Wang et al., we pair this test with random-head controls and a utility check, which makes the trade-off visible.

Memorization, extraction, and distributed leakage. A separate literature argues against overly local explanations for leakage. Deduplicated training data greatly reduces memorized emissions while improving efficiency, which shows that leakage can be amplified by data duplication rather than by one specific circuit Lee et al. [2022]. Strong extraction baselines further show that leakage depends on prompting and search strategy, not just on the presence of a memorized string Yu et al. [2023]. Ippolito et al. show that blocking exact verbatim memorization does not eliminate leakage under modified prompts Ippolito et al. [2022]. Huang et al. go further and argue that verbatim memorization is intertwined with general language modeling capability, making it hard to isolate without utility loss Huang et al. [2024]. Our results align more closely with this distributed view than with a single-circuit story.

Diffuse distributions and random output failures. Zhang et al. show that instruction-tuned LLMs are systematically poor at producing diffuse distributions when prompts ask for random answers Zhang et al. [2024]. That work motivates our random-choice leakage task: if models already struggle to be diffuse, repeated prompt context may further bias them toward copying salient items. Unlike Zhang et al., we do not train a mitigation. Instead, we use mechanistic interventions to ask whether induction-like heads are a major cause of that failure mode.

Positioning. Our study sits at the intersection of these themes. We combine empirical induction-head detection, causal ablation, a random-choice leakage metric, and a utility control in one real model. The result is a more constrained claim than some mechanistic accounts suggest: induction-like heads matter, but their effect on leakage-like copying is heterogeneous and not cleanly separable from broader model behavior.

3 Methodology

Research question. We ask whether induction-like heads in GPT-2 SMALL materially increase unintended copying pressure on prompts that should instead produce diffuse or random outputs. The experiment is designed to connect head-level interventions to two behaviors in the same model: intended copying on a standard induction-style continuation task and unintended copying on a constrained random-choice task.

3.1 Model, toolkit, and compute

We use GPT-2 SMALL with TransformerLens, which gives direct access to attention heads and head-level activation hooks. The local environment used PyTorch 2.11.0+cu130, Transformers 5.5.4, Datasets 4.8.4, SciPy 1.17.1, Pandas 3.0.2, Matplotlib 3.10.8, and Seaborn 0.13.2. The machine exposed four NVIDIA RTX A6000 GPUs through `nvidia-smi`, but PyTorch could not use CUDA because the installed driver-runtime pair was incompatible with the local torch build. All runs therefore executed on CPU. This constrained sample sizes but kept the intervention protocol identical across conditions.

3.2 Head identification

We identify candidate induction heads using synthetic repeated-token prompts over six single-token color candidates: `red`, `blue`, `green`, `yellow`, `black`, and `white`. For each head, we compute TransformerLens's `detect_head(..., "induction_head")` score across 48 head-detection prompts in the final run. The top two heads are `L5H5` and `L5H1` with scores 0.740 and 0.548, respectively. Figure 1 shows the resulting score profile, which places the highest-scoring heads in the familiar mid-layer region reported by prior work Olsson et al. [2022], Crosbie and Shutova [2024].

3.3 Tasks

Copy task. The copy task uses prompts of the form `A B C D | A B C` and evaluates whether the model predicts `D` next. This is a direct induction-style continuation pattern. We evaluate 96 prompts in the final run and measure top-1 accuracy and mean target logit.

Random-choice leakage task. The random-choice task asks the model to choose one item from the same six-item candidate set with wording such as "with no preference" or "at random." We compare a neutral prompt against a matched prompt that additionally includes repeated distractor context, such as `Previous answers: red, red, red`. For each of 96 prompts, we measure the probability assigned to the repeated distractor in both variants.

Our primary leakage metric is the repeated-token bias delta

$$\Delta_{\text{bias}} = P(\text{repeated} \mid \text{biased prompt}) - P(\text{repeated} \mid \text{neutral prompt}). \tag{1}$$

Higher values indicate stronger prompt-induced pressure toward the repeated distractor. We also compute normalized entropy over the six valid candidates,

$$H_{\text{norm}} = \frac{-\sum_{i=1}^{6} p_i \log p_i}{\log 6}, \tag{2}$$

Table 1: Main final-run results. Lower is better for repeated-token bias delta, KL to uniform, and WikiText perplexity. Higher is better for normalized entropy when the goal is a diffuse distribution. Best values are shown in **bold**.

Condition	Copy accuracy	Bias delta	Normalized entropy	KL to uniform	WikiText perplexity
Full model	0.729	0.0496	**0.860**	**0.152**	**89.9**
Induction ablation (L5H5 and L5H1)	**0.823**	**0.0211**	0.819	0.227	169.7
Random-head ablation (mean)	0.723	0.0466	0.847	0.175	106.5

where larger values indicate a more diffuse distribution, and KL divergence to the uniform distribution as a secondary diagnostic.

Utility check. To measure whether the intervention preserves general language modeling quality, we compute negative log-likelihood and perplexity on 12 non-empty samples from WIKITEXT-103 using up to 64 tokens per sample.

3.4 Interventions and baselines

We compare three conditions:

- **Full model:** no intervention.
- **Induction-head ablation:** zero the selected heads' `hook_z` activations in TransformerLens.
- **Random-head ablation control:** zero a matched number of randomly chosen heads and average across five random head sets.

The main final run uses top-2 induction-like heads, while sensitivity runs test two alternate high-scoring subsets: `L5H5,L6H9` and `L5H5,L6H9,L5H1`. The random-head control is important because hard head ablation can change model behavior for many generic reasons unrelated to induction-like circuitry.

3.5 Data resources

We use the local `datasets/synthetic_copy_task/` resource to validate the availability of a copy-style benchmark and to ground the experimental setup. We also use the local `datasets/wikitext_103_v1/` resource for the utility check. The final report records 900,000 synthetic-copy training rows and 1,801,350 WikiText training rows in the local environment.

3.6 Statistical analysis

The final run uses paired prompt-level comparisons for the random-choice task. Because Shapiro-Wilk tests on the paired differences rejected normality, we report Wilcoxon signed-rank tests, 95% bootstrap confidence intervals for mean differences, and Benjamini-Hochberg adjusted p-values over the main comparison family. We report paired effect sizes as Cohen's d. This analysis focuses on two questions: whether induction-head ablation differs from the full model and whether it differs from the random-head control.

4 Results

4.1 Detected induction-like heads match canonical mid-layer structure

The induction-head detector identifies a small number of clearly induction-like heads, with `L5H5` and `L5H1` scoring 0.740 and 0.548 in the final run. This localization is important because it shows that the intervention targets a plausible induction circuit rather than an arbitrary collection of heads.

4.2 Main results: bias falls, but entropy and utility fall with it

Table 1 summarizes the main final run. The strongest induction-head ablation lowers repeated-token bias delta from 0.0496 in the full model to 0.0211, a decrease of 57.5%. Relative to the

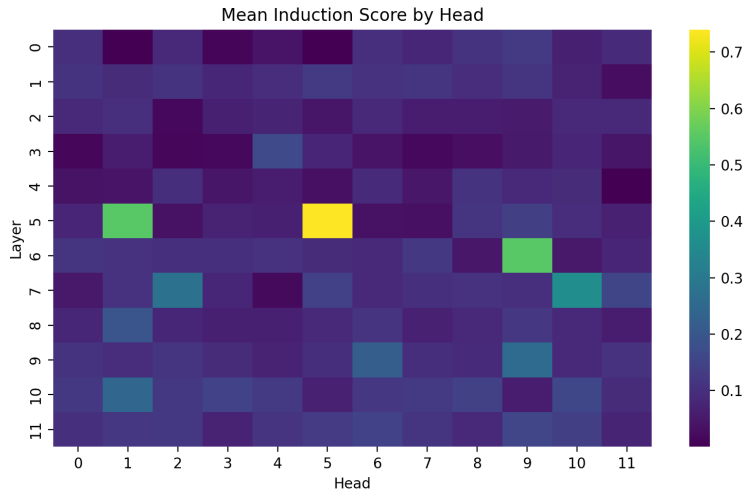

Figure 1: Induction-head detection scores across attention heads in GPT-2 SMALL. The highest-scoring heads are concentrated in mid layers, with `L5H5` and `L5H1` emerging as the top two heads in the final run. This matches the qualitative pattern reported in prior induction-head studies.

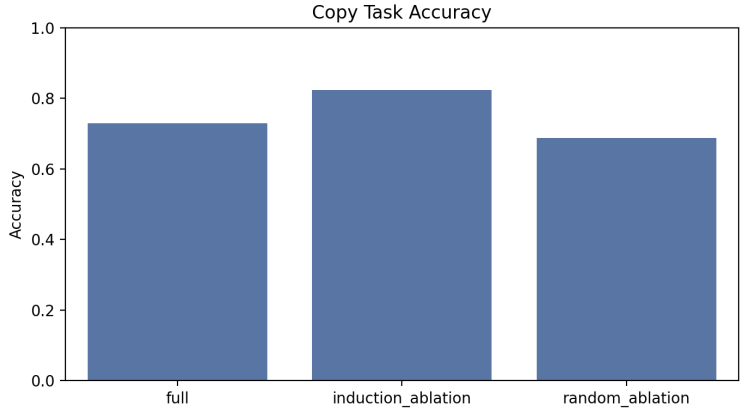

Figure 2: Copy-task accuracy across full, induction-ablation, and random-head conditions. The final run does not show a clean "remove induction heads, reduce copying" pattern: ablating `L5H5` and `L5H1` increases copy-task accuracy above both the full model and the mean random-head control.

mean random-head control (0.0466), the intervention remains clearly lower. However, the same ablation also lowers normalized entropy from 0.860 to 0.819 and raises WikiText perplexity from 89.9 to 169.7. The random-head control also hurts utility, but much less severely, reaching a mean perplexity of 106.5.

The copy task adds an important complication. Ablating `L5H5` and `L5H1` *increases* copy-task top-1 accuracy from 0.729 to 0.823 rather than lowering it. This means the top-scoring induction-like subset in the final run is not behaving like a clean copy-only circuit. The intervention changes behavior, but not in the simple direction implied by the original hypothesis.

4.3 Prompt-level tests show a real effect, but not a clean mitigation

Table 2 reports paired prompt-level comparisons for the random-choice task. All four primary comparisons are significant after Benjamini-Hochberg correction. Relative to the full model, induction-head ablation lowers repeated-token bias by 0.0285 (95% CI $[-0.0400, -0.0172]$, Wilcoxon $p = 1.54 \times 10^{-7}$) but also lowers normalized entropy by 0.0417 (95% CI $[-0.0552, -0.0274]$,

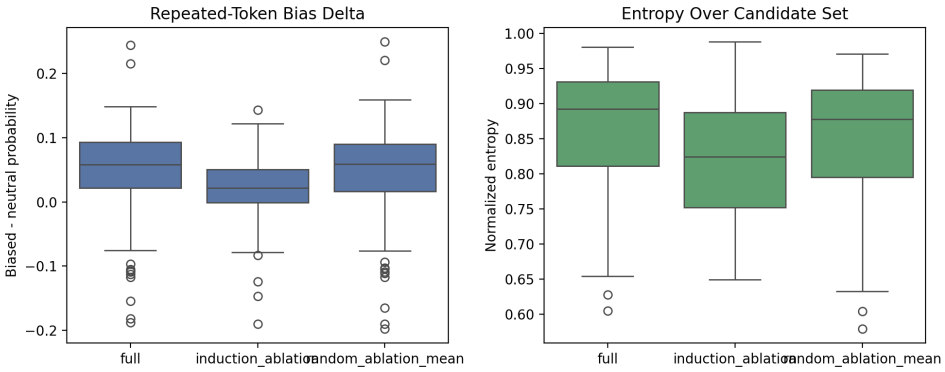

Figure 3: Random-choice leakage metrics across conditions. Induction-head ablation lowers repeated-token bias but also lowers entropy over the valid candidate set. Lower bias therefore does not translate into more diffuse output distributions.

Table 2: Prompt-level paired comparisons for the final random-choice run. Mean difference is condition A minus condition B. All tests use Wilcoxon signed-rank because the paired differences are non-normal.

Comparison	Mean diff	95% bootstrap CI	Test	p	BH-adjusted p	Effect size
Bias delta: induction vs. full	-0.0285	$[-0.0400, -0.0172]$	Wilcoxon	1.54×10^{-7}	4.23×10^{-7}	-0.51
Bias delta: induction vs. random	-0.0255	$[-0.0370, -0.0145]$	Wilcoxon	8.08×10^{-7}	1.08×10^{-6}	-0.46
Entropy: induction vs. full	-0.0417	$[-0.0552, -0.0274]$	Wilcoxon	2.11×10^{-7}	4.23×10^{-7}	-0.59
Entropy: induction vs. random	-0.0289	$[-0.0430, -0.0142]$	Wilcoxon	2.21×10^{-5}	2.21×10^{-5}	-0.40

Wilcoxon $p = 2.11 \times 10^{-7}$). The comparison against the random-head control shows the same pattern: lower bias and lower entropy.

These prompt-level results matter because they rule out the simplest null hypothesis that the final-run reduction in repeated-token bias is just noise. At the same time, they also rule out the strongest positive interpretation. The intervention does not make the model more uniformly random over the valid choices. It sharpens the distribution in a different way.

4.4 Sensitivity runs flip the causal story

Table 3 is the most important result in the paper. In the smaller top-2 sensitivity run, ablating `L5H5,L6H9` drops copy accuracy from 0.542 to 0.250 but *increases* repeated-token bias from 0.0371 to 0.0804. Adding `L5H1` in the top-3 run still leaves bias above the full model at 0.0638 and drives perplexity to 201.45. The final run with `L5H5` and `L5H1` instead lowers bias to 0.0211 while also increasing copy accuracy.

This sign flip means that "induction-like heads" are not interchangeable for the purpose of leakage mitigation. Different high-scoring subsets push the model in different directions. A head can look induction-like under the detector while still playing a heterogeneous role once we intervene on actual behavior.

5 Discussion

What do the results show? The data support a narrow but useful claim: induction-like heads in GPT-2 SMALL do affect repeated-token bias on prompts that ask for a random answer. The data do *not* support the stronger claim that induction heads are a large singular source of leakage. The best evidence against that stronger claim is the sensitivity analysis. Depending on which high-scoring heads we ablate, repeated-token bias can either fall or rise. That is hard to reconcile with a monolithic circuit explanation.

Why does the final run still matter? The final run shows a statistically strong reduction in repeated-token bias, and that effect survives comparison to the random-head control. If we only looked at that

Table 3: Sensitivity runs over different high-scoring induction-like head subsets. The sign of the repeated-token bias effect changes across subsets, which is the main evidence against a monolithic leakage-circuit interpretation.

Run	Heads	Copy acc. full	Copy acc. ablated	Bias full	Bias ablated	Entropy full	Entropy ablated	WikiText ppl ablated
Sweep top-2	`L5H5,L6H9`	0.542	0.250	0.0371	0.0804	0.8701	**0.8720**	**86.83**
Sweep top-3	`L5H5,L6H9,L5H1`	0.542	0.333	0.0371	0.0638	0.8701	0.7909	201.45
Final run	`L5H5,L5H1`	**0.729**	**0.823**	0.0496	**0.0211**	0.8602	0.8186	169.74

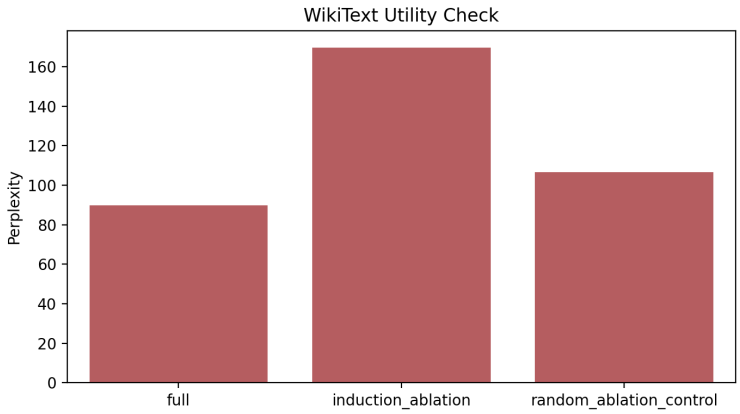

Figure 4: WikiText perplexity under the main intervention and random-head control. Hard ablation of the detected induction-like heads substantially harms language modeling utility, which makes the intervention unattractive as a leakage-specific fix.

run, we might conclude that induction heads are a promising mitigation target. The problem is that the same intervention reduces entropy and harms WikiText utility, while nearby head subsets reverse the direction of the bias effect. The result is therefore real but not robust enough to justify a clean mechanistic story.

How should we interpret the entropy drop? Lower repeated-token bias is not the same as better randomness. In our setup, ablating `L5H5` and `L5H1` reduces the probability assigned to the repeated distractor, but it also makes the candidate distribution less diffuse overall. That pattern echoes the broader randomness-control literature Zhang et al. [2024]: moving one undesired mode down does not guarantee a more uniform distribution across the valid options. Mechanistic mitigation needs to track distributional quality directly, not only the probability of one repeated token.

Connection to memorization and leakage work. Our results fit better with the literature that treats memorization and leakage as distributed phenomena Lee et al. [2022], Ippolito et al. [2022], Huang et al. [2024]. Hard ablation can change one observable behavior, but the effect is entangled with general model quality and with the interaction among heads. That does not make mechanistic analysis unhelpful. It means the analysis should aim for richer behavior-specific interventions, such as softer descaling Wang et al. [2025], rather than assuming one clean deletion will isolate leakage.

Limitations. This study tests only one model family, GPT-2 SMALL. The random-choice task is synthetic and uses a base model rather than an instruction-tuned model, so the prompt instruction to choose randomly is weaker than it would be in a chat model. CPU-only execution limited the final sample sizes and the breadth of sensitivity sweeps. The utility check uses only 12 WikiText samples, which is enough to show a large degradation but not enough to characterize subtle language modeling effects. Finally, we do not run a true training-data extraction attack, so our "leakage" metric should be read as prompt-induced copying pressure rather than direct memorization extraction.

Broader implications. The main practical lesson is methodological. Circuit-level interventions for leakage should not be evaluated on a single synthetic metric in isolation. They need at least three checks: whether the target behavior changes, whether the output distribution remains well-formed, and whether general utility survives. Our results show that a promising-looking mechanistic effect can fail that broader evaluation.

6 Conclusion

We presented a head-level causal study of whether induction-like heads are a large source of leakage in GPT-2 SMALL. Using TransformerLens, we identified high-scoring induction-like heads, ablated them on both copy and random-choice tasks, and compared the effect to random-head controls and a WikiText utility check.

The core finding is not a clean mitigation result but a sensitivity result. Ablating `L5H5` and `L5H1` reduces repeated-token bias on random-choice prompts, yet it also lowers entropy and sharply harms utility. Nearby high-scoring head subsets reverse the bias effect entirely. The strongest supported conclusion is therefore narrow: induction-like heads modulate repeated-token bias, but they are not a singular or stable source of leakage-like copying in this model.

The next step is to test the same protocol where "choose randomly" is behaviorally meaningful, to replace hard zero ablation with softer descaling, and to pair these interventions with an extraction benchmark. That combination would provide a stronger answer about whether mechanistic copying circuits can be mitigated without collapsing useful language modeling behavior.

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